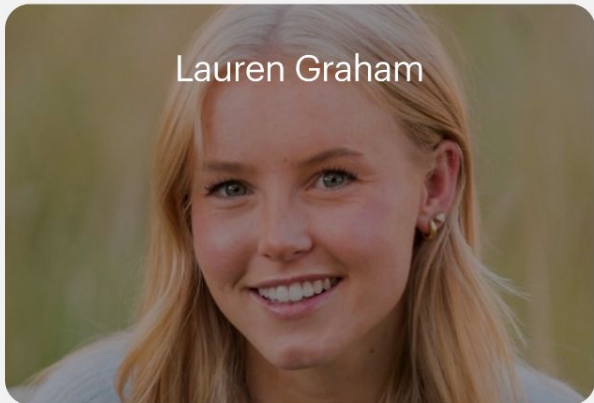


# Støljin

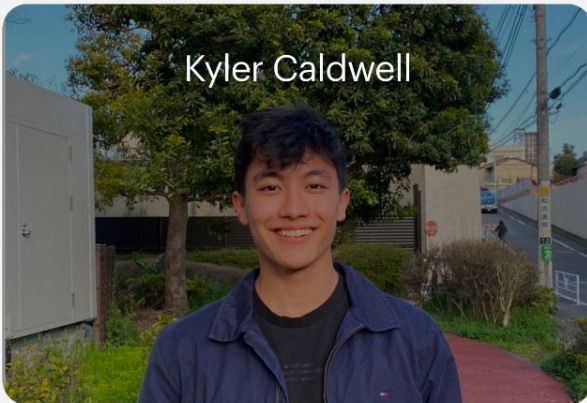
Personalized seating for  
individuals with herniated discs

# Our team.

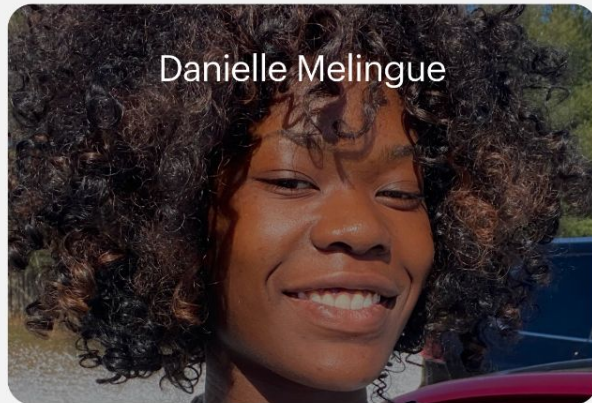
Lauren Graham



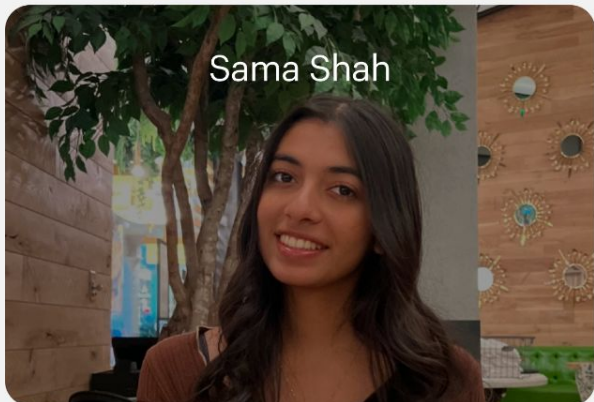
Kyler Caldwell



Danielle Melingue



Sama Shah



Byeongjun Moon



Hamin Jin



# My dad suffered from a slip herniated disc in Korea while working his job.



Hamin Jin (left) and Mr. Jin (right)

And this year, my sister also started to suffer from extreme back pain, which is also most likely a herniated disc.

Both of them, even today, still live with lingering effects from the pain of herniated discs.

# Here's some more research on slip-herniated disc.

160m+

people get a herniated disc every year

30-50

age range of people most likely to get herniated discs

3.18hr

average hours of civilian workers spent sitting a day

8hr

for the 90th percentile

The main risk factor is sitting for long periods in the same position.

# Our inclusive design process.

To begin, we asked some questions.

What types of chairs typically cause you discomfort?

How long do you usually sit before becoming uncomfortable?

What solutions have you applied to this situation?

Are there any solutions you recommend or have thought about?

How can we make a solution more accessible for you?



And here are the feedback we received.

*Unsuitable for  
long hours*

All while practicing inclusive processes, such as:

- consulting with our group in between design drafts and prototyping
- incorporating ergonomic recommendations such as proper lumbar support, adjustable backrest angles and comfortable seating
- consulting with healthcare professionals and experts to ensure our design are a safe and sustainable long term solution

Complicated

Lumbar Support

Expensive

# Our design theory.

Sitting with herniated disc means making sure that you sit with perfect posture.

A natural aesthetic, inspired by nature.

## Perfect back position

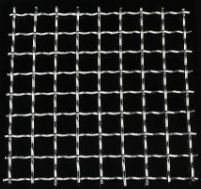
- Shoulders back
- Arched lower back support
- Knees below waist level



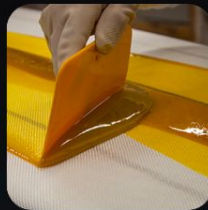


# Engineering and design.

Customizable design inspired by airplane headrests, where the cushion is moldable around your neck.

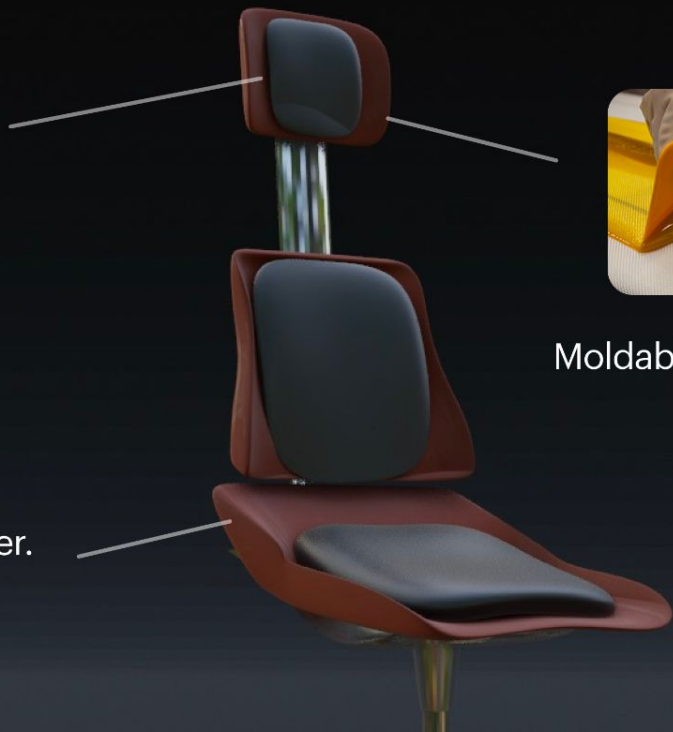


Wire mesh frame



Moldable fiberglass

Wood, leather, or carbon fiber.  
Your choice.



# Accessible for broader demographics.

All



of muscular usage across  
all sports

20%



of sports injuries involve  
lower back

75%



of 60+ yr olds with back pain after  
sitting for 2+ hrs

70%



experience lower back pain  
when sitting



Thank you.